From Predictive Models to Bayesian Functions · Quiz explanations

Original W7 formative checkpoints aligned with the main note; model identities follow its textbook reading map.

1. Does drawing more posterior functions give us more observed training examples?

No. The observed data remain fixed. Draws represent plausible functions and are used to calculate posterior quantities.

2. Under which condition does Gaussian negative log likelihood select the same mean parameters as squared error?

For a common fixed positive observation variance, with the same mean model and data. Variance-dependent terms cannot be discarded when comparing or learning changing variances.

3. In logistic regression, which function is the link and which is the inverse link?

The logit maps the Bernoulli mean probability to a real score and is the link. The sigmoid maps the score back to a probability and is the inverse link.

4. Does running the same deterministic fitted network repeatedly produce posterior samples?

No. Fixed inputs and fixed weights return the same result. Posterior sampling requires an inference procedure for a parameter or function distribution.

5. If we keep the data and posterior fixed but use more Monte Carlo draws, which uncertainty should decrease?

The numerical error of a suitable Monte Carlo estimate typically decreases. The posterior spread and observation-noise variance are not reduced by additional computation alone.

6. Why is the interval for a future noisy observation wider than the interval for its latent function value?

Under independent additive Gaussian noise, future-observation variance equals latent-function variance plus the positive observation-noise variance. Both intervals have the same conditional mean.

7. What must match when comparing weight-space and function-space predictions?

The feature map, prior mean and covariance, noise model, observed data and prediction inputs. The kernel must be induced by that same feature prior.

8. Can a Bayesian model have narrow intervals and poor coverage? Explain one reason.

Yes. A misspecified function family, likelihood, prior or fixed hyperparameter choice can produce confident but systematically wrong predictions. Integrating weights does not validate those assumptions.